



第10章 二端口网络

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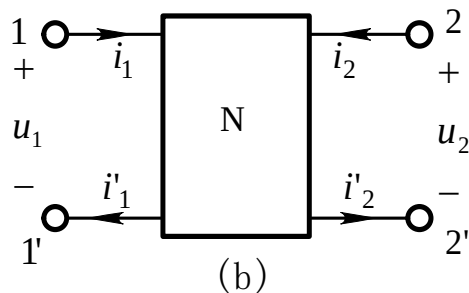
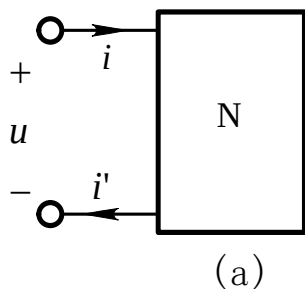
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§ 10.1 二端口网络

1 二端口网络:



端口变量:

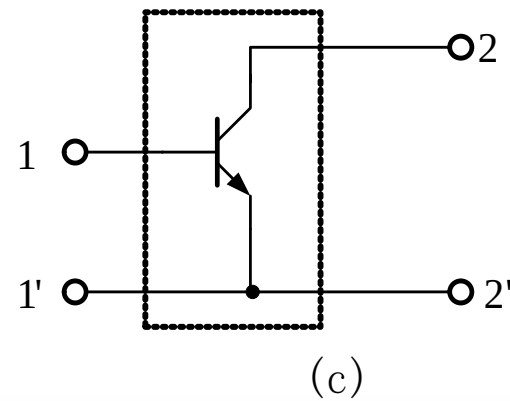
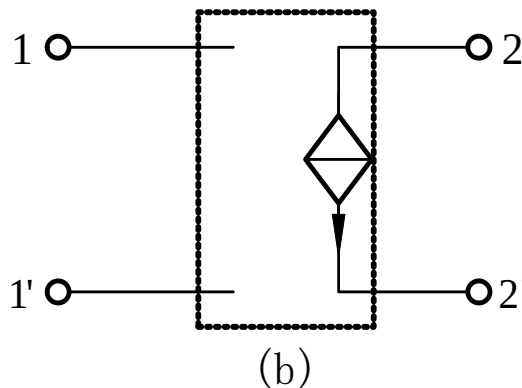
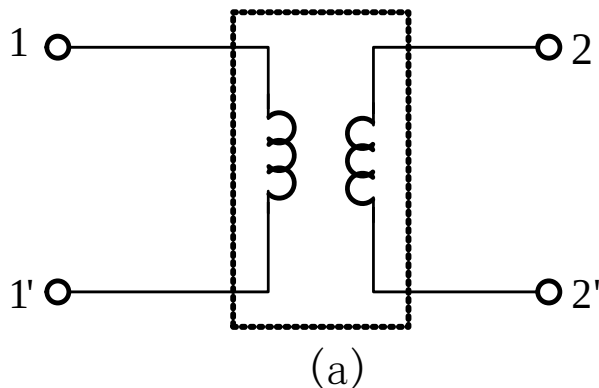
$$U_1, I_1, U_2, I_2;$$

$$u_1, i_1, u_2, i_2;$$

$$\dot{U}_1, \dot{I}_1, \dot{U}_2, \dot{I}_2;$$

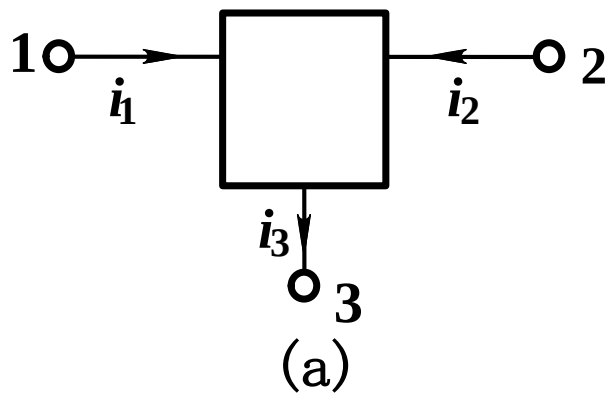
$$U_1(s), I_1(s), U_2(s), I_2(s)$$

二端口网络举例



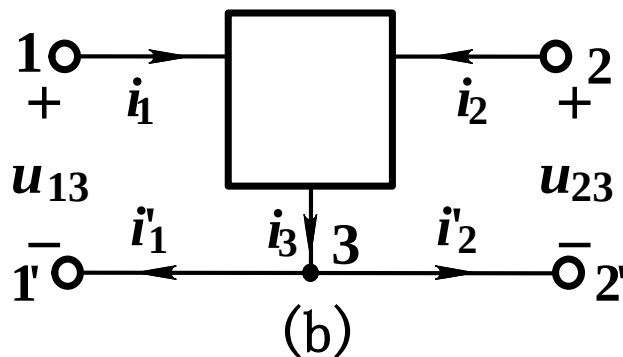
§ 10.1 二端口网络

2 用二端口等效代替三端网络



$$i_3 = i_1 + i_2$$

$$u_{12} = u_{13} - u_{23}$$



三端网络只须用两个独立的端子电流和两个独立的端子间电压来描述如 (b)

推广：一个n端网络可用n-1端口等效代替

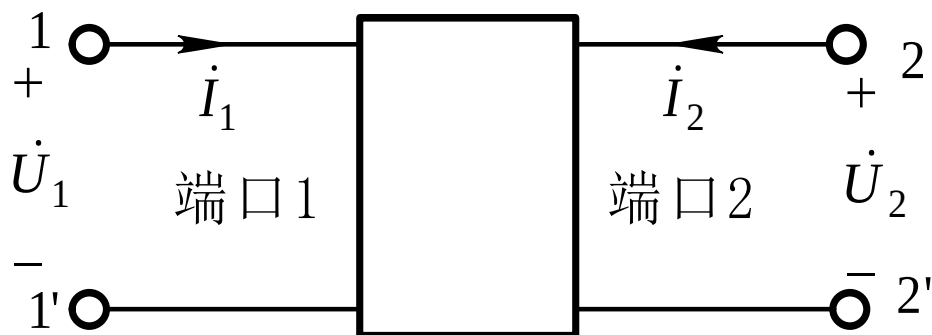
3 讨论范围

含线性 R 、 L 、 C 、 M 与线性受控源

不含独立源(复频域分析时, 不包含附加电源)。

§ 10.2 导纳参数方程和阻抗参数方程

1 导纳参数方程



导纳参数方程或Y参数方程

$$\dot{I}_1 = Y_{11}\dot{U}_1 + Y_{12}\dot{U}_2$$

$$\dot{I}_2 = Y_{21}\dot{U}_1 + Y_{22}\dot{U}_2$$

矩阵形式:

$$\begin{bmatrix} \dot{I}_1 \\ \dot{I}_2 \end{bmatrix} = \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} \begin{bmatrix} \dot{U}_1 \\ \dot{U}_2 \end{bmatrix}$$

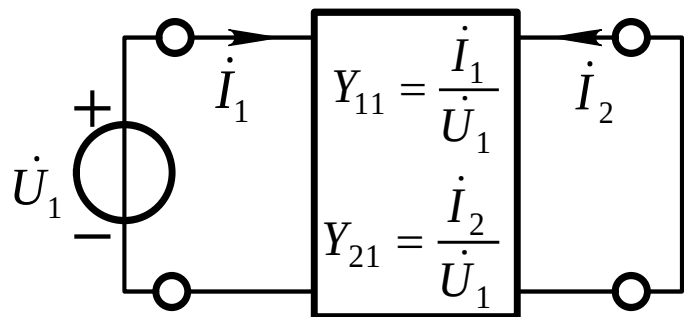
向量形式: $\dot{\mathbf{I}} = \mathbf{Y}\dot{\mathbf{U}}$

式中: $\dot{\mathbf{U}} = [\dot{U}_1, \dot{U}_2]^T$ 和 $\dot{\mathbf{I}} = [\dot{I}_1, \dot{I}_2]^T$ 分别表示端口电压和电流向量

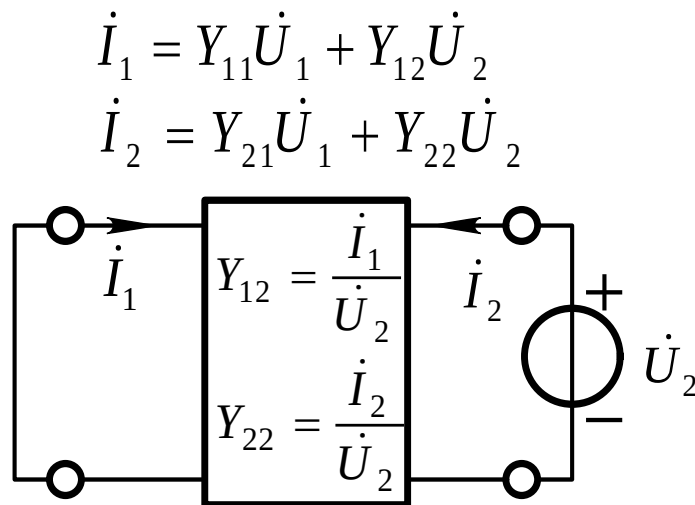
$$\mathbf{Y} = \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} \quad (\text{导纳参数矩阵或Y参数矩阵})$$

§ 10.2 导纳参数方程和阻抗参数方程

导纳参数的测定



(a)



(b)

$$Y_{11} = \left. \frac{\dot{I}_1}{\dot{U}_1} \right|_{\dot{U}_2=0}$$

$$Y_{21} = \left. \frac{\dot{I}_2}{\dot{U}_1} \right|_{\dot{U}_2=0}$$

$$Y_{12} = \left. \frac{\dot{I}_1}{\dot{U}_2} \right|_{\dot{U}_1=0}$$

$$Y_{22} = \left. \frac{\dot{I}_2}{\dot{U}_2} \right|_{\dot{U}_1=0}$$

Y参数也称为短路导纳参数

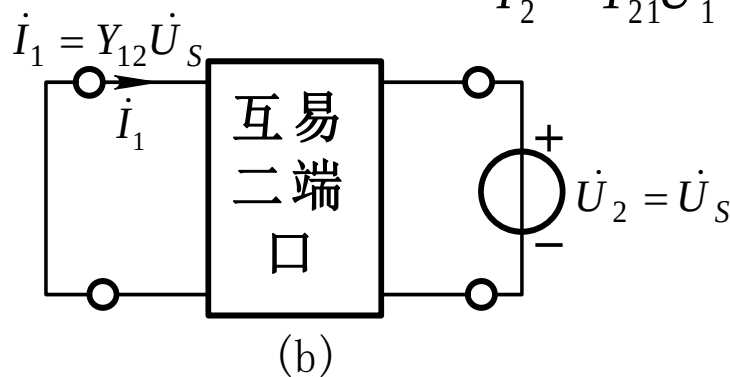
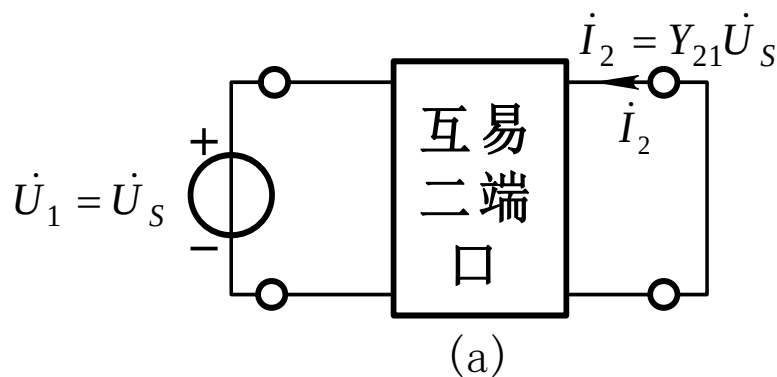
Y_{11} -- 短路输入导纳

Y_{22} -- 短路输出导纳

Y_{12} Y_{21} -- 短路转移导纳

§ 10.2 导纳参数方程和阻抗参数方程

互易及对称情况：



$$\begin{aligned}\dot{I}_1 &= Y_{11}\dot{U}_1 + Y_{12}\dot{U}_2 \\ \dot{I}_2 &= Y_{21}\dot{U}_1 + Y_{22}\dot{U}_2\end{aligned}$$

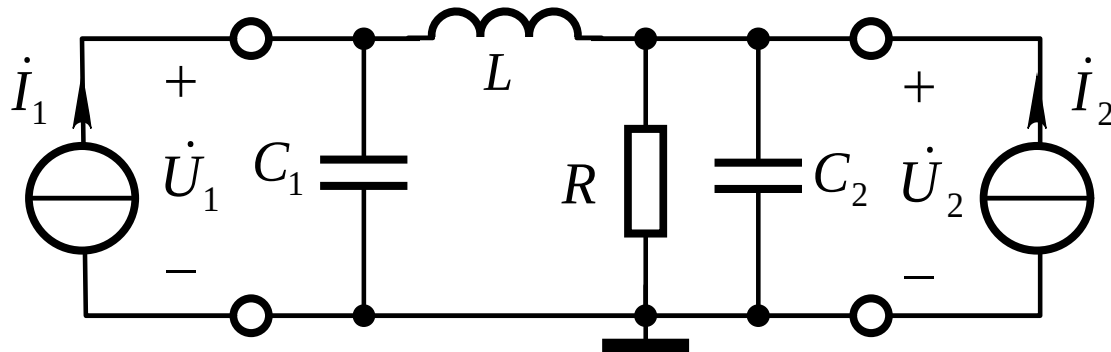
图中互易二端口的电流满足 $\dot{I}_2 = \dot{I}_1$ $\dot{I}_1 = Y_{12}\dot{U}_S$ $\dot{I}_2 = Y_{21}\dot{U}_S$

$\Rightarrow Y_{12} = Y_{21}$

反之如果 Y 参数满足 $Y_{12} = Y_{21}$ 则此二端口是互易二端口。

如果同时满足 $Y_{12} = Y_{21}$ 和 $Y_{11} = Y_{22}$ 则称为对称二端口。

§ 10.2 导纳参数方程和阻抗参数方程



例 求左图所示二端口Y参数矩阵。

解

用电流源置换两个端口列节点电压方程

$$\dot{I}_1 = (j\omega C_1 + \frac{1}{j\omega L})\dot{U}_1 - \frac{1}{j\omega L}\dot{U}_2$$

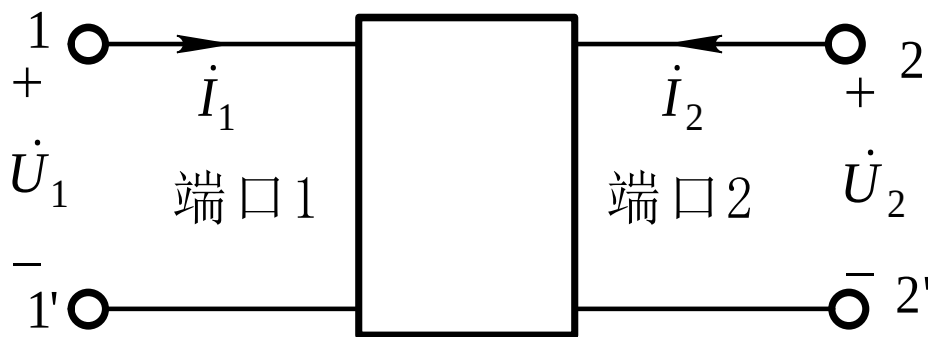
$$\dot{I}_2 = -\frac{1}{j\omega L}\dot{U}_1 + (\frac{1}{R} + j\omega C_2 + \frac{1}{j\omega L})\dot{U}_2$$

上式的系数矩阵就是所求Y参数矩阵：

$$\mathbf{Y} = \begin{bmatrix} j(\omega C_1 - \frac{1}{\omega L}) & -\frac{1}{j\omega L} \\ -\frac{1}{j\omega L} & \frac{1}{R} + j(\omega C_2 - \frac{1}{\omega L}) \end{bmatrix}$$

§ 10.2 导纳参数方程和阻抗参数方程

2 阻抗参数方程



阻抗参数方程或Z参数方程

$$\begin{aligned}\dot{U}_1 &= Z_{11}\dot{I}_1 + Z_{12}\dot{I}_2 \\ \dot{U}_2 &= Z_{21}\dot{I}_1 + Z_{22}\dot{I}_2\end{aligned}$$

矩阵形式:
$$\begin{bmatrix} \dot{U}_1 \\ \dot{U}_2 \end{bmatrix} = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} \begin{bmatrix} \dot{I}_1 \\ \dot{I}_2 \end{bmatrix}$$

向量形式:
$$\dot{U} = \mathbf{Z}\dot{I}$$

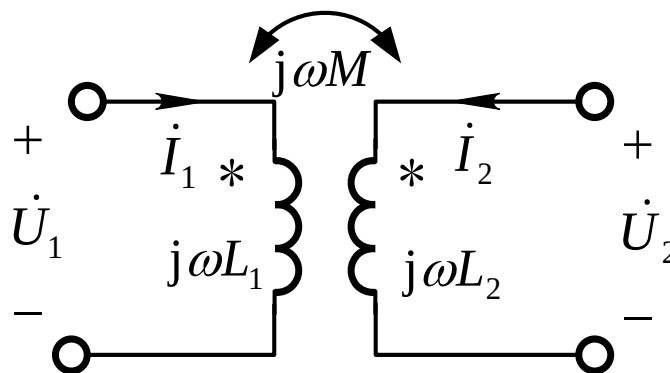
$$\mathbf{Z} = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix}$$
 二端口阻抗参数矩阵
或Z 参数矩阵

• 互易条件: $Z_{12} = Z_{21}$

• 对称条件: $Z_{12} = Z_{21}$ 和 $Z_{11} = Z_{22}$

§ 10.2 导纳参数方程和阻抗参数方程

例：如图所示二端口互感：



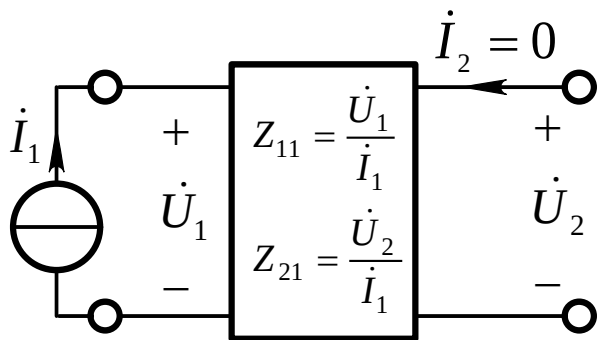
$$\begin{bmatrix} \dot{U}_1 \\ \dot{U}_2 \end{bmatrix} = \begin{bmatrix} j\omega L_1 & j\omega M \\ j\omega M & j\omega L_2 \end{bmatrix} \begin{bmatrix} \dot{I}_1 \\ \dot{I}_2 \end{bmatrix}$$

$$\mathbf{Z} = \begin{bmatrix} j\omega L_1 & j\omega M \\ j\omega M & j\omega L_2 \end{bmatrix}$$

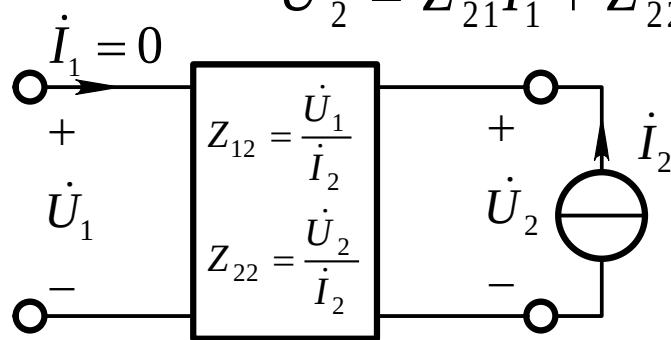
§ 10.2 导纳参数方程和阻抗参数方程

二端口的Z参数测定:

$$\begin{aligned}\dot{U}_1 &= Z_{11}\dot{I}_1 + Z_{12}\dot{I}_2 \\ \dot{U}_2 &= Z_{21}\dot{I}_1 + Z_{22}\dot{I}_2\end{aligned}$$



(a)



(b)

$$Z_{11} = \left. \frac{\dot{U}_1}{\dot{I}_1} \right|_{\dot{I}_2=0}$$

$$Z_{21} = \left. \frac{\dot{U}_2}{\dot{I}_1} \right|_{\dot{I}_2=0}$$

$$Z_{12} = \left. \frac{\dot{U}_1}{\dot{I}_2} \right|_{\dot{I}_1=0}$$

$$Z_{22} = \left. \frac{\dot{U}_2}{\dot{I}_2} \right|_{\dot{I}_1=0}$$

Z参数也称为**开路阻抗参数**

Z_{11} -- 开路输入阻抗

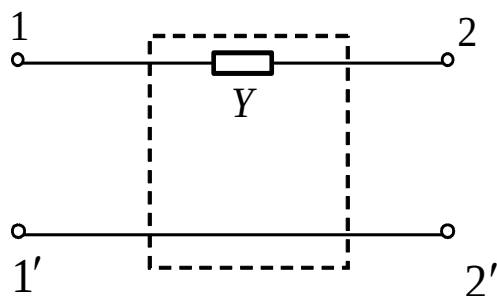
Z_{22} -- 开路输出阻抗

Z_{12} Z_{21} -- 开路转移阻抗

§ 10.2 导纳参数方程和阻抗参数方程

特殊情况

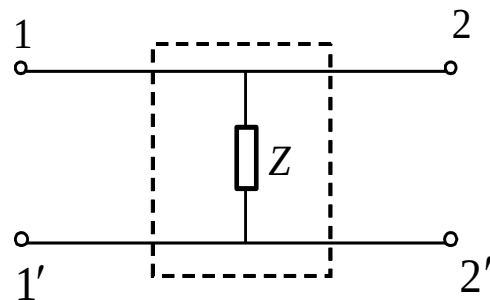
对于给定的二端口，有时不同时存在阻抗参数矩阵和导纳参数矩阵



简单串联元件组成的二端口

图中没有阻抗参数矩阵，
只有导纳参数矩阵

$$\mathbf{Y} = \begin{bmatrix} Y & -Y \\ -Y & Y \end{bmatrix}$$



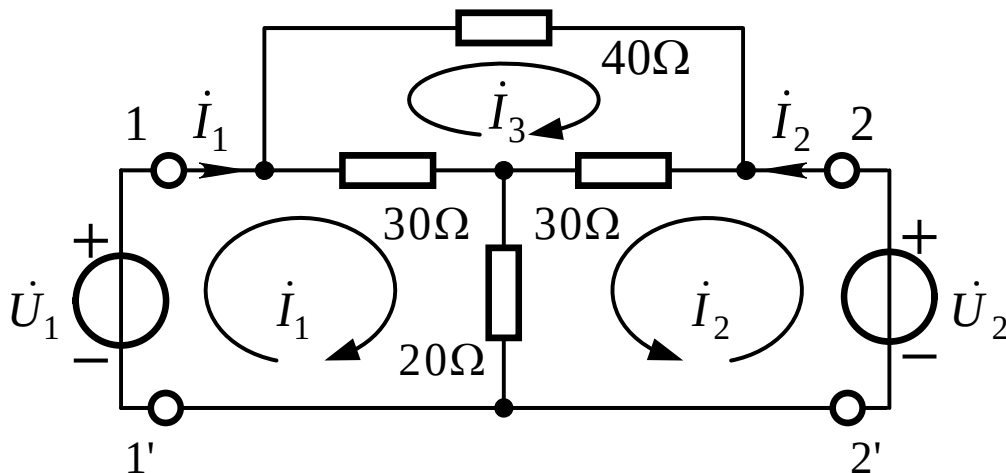
简单并联元件组成的二端口

图中没有导纳参数矩阵，
只有阻抗参数矩阵：

$$\mathbf{Z} = \begin{bmatrix} Z & Z \\ Z & Z \end{bmatrix}$$

§ 10.2 导纳参数方程和阻抗参数方程

例 求如图所示二端口的阻抗参数矩阵。



解

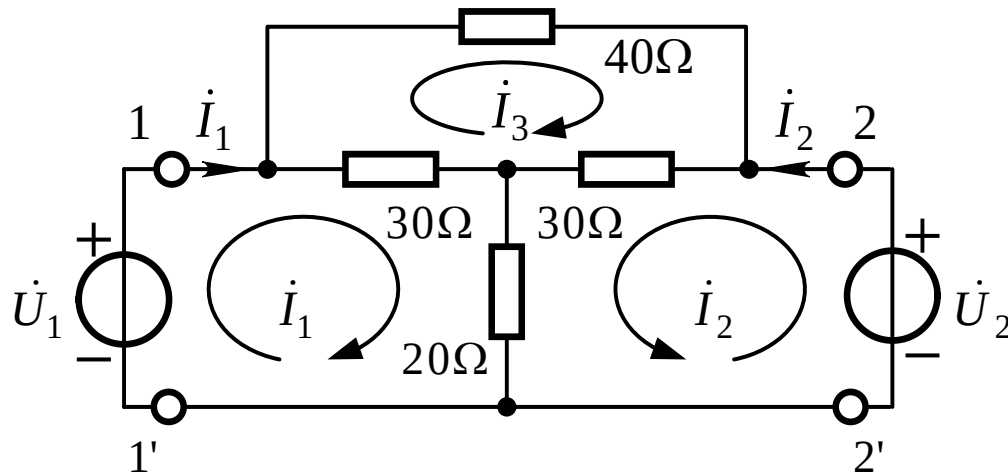
求 Z 参数宜列回路电流方程。用电压源 $\dot{U}_1$ 和 $\dot{U}_2$ 分别置换端口1和端口2的外接电路。

$$(30 + 20)\Omega \times \dot{I}_1 + 20\Omega \times \dot{I}_2 - 30\Omega \times \dot{I}_3 = \dot{U}_1 \quad (1)$$

$$20\Omega \times \dot{I}_1 + (30 + 20)\Omega \times \dot{I}_2 + 30\Omega \times \dot{I}_3 = \dot{U}_2 \quad (2)$$

$$-30\Omega \times \dot{I}_1 + 30\Omega \times \dot{I}_2 + (30 + 30 + 40)\Omega \times \dot{I}_3 = 0 \quad (3)$$

§ 10.2 导纳参数方程和阻抗参数方程



由方程(3)解出 $\dot{I}_3$ ，再代入方程(1)、(2)：

$$41\Omega \times \dot{I}_1 + 29\Omega \times \dot{I}_2 = \dot{U}_1$$

$$29\Omega \times \dot{I}_1 + 41\Omega \times \dot{I}_2 = \dot{U}_2$$

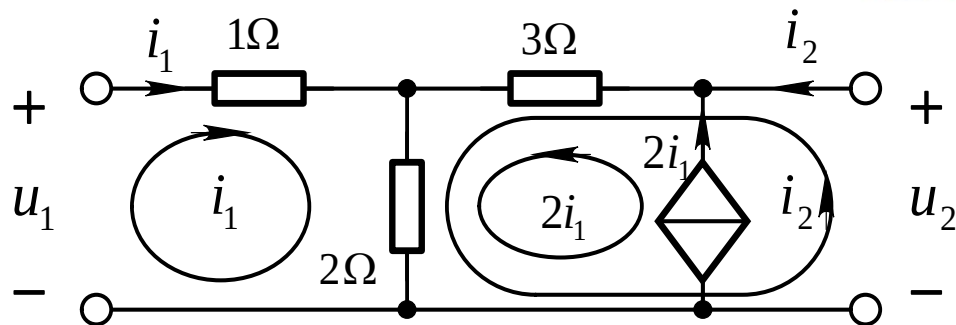
$$Z\text{参数矩阵为 } \mathbf{Z} = \begin{bmatrix} 41 & 29 \\ 29 & 41 \end{bmatrix} \Omega$$

本题电路结构与元件参数对称是对称二端口，故

$$Z_{12} = Z_{21}, Z_{11} = Z_{22}$$

§ 10.2 导纳参数方程和阻抗参数方程

例 求图所示二端口的阻抗参数矩阵。



解 选图中回路列回路电流方程

$$(1 + 2)\Omega \times i_1 + 2\Omega \times i_2 + 2\Omega \times 2i_1 = u_1$$

$$2\Omega \times i_1 + (2 + 3)\Omega \times 2i_1 + (2 + 3)\Omega \times i_2 = u_2$$

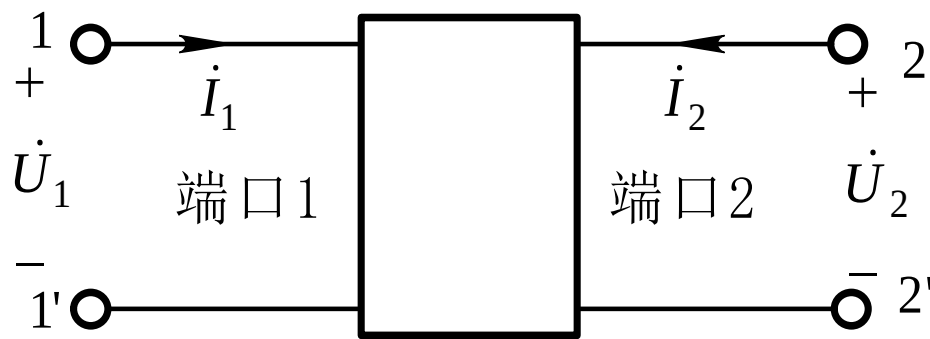
整理得： $7\Omega \times i_1 + 2\Omega \times i_2 = u_1$

$$12\Omega \times i_1 + 5\Omega \times i_2 = u_2$$

Z参数矩阵为 $\mathbf{Z} = \begin{bmatrix} 7 & 2 \\ 12 & 5 \end{bmatrix} \Omega$

§ 10.3 传输参数和混合参数

1 传输参数方程



传输参数方程或A参数方程

$$\begin{aligned}\dot{U}_1 &= A_{11}\dot{U}_2 + A_{12}(-\dot{I}_2) \\ \dot{I}_1 &= A_{21}\dot{U}_2 + A_{22}(-\dot{I}_2)\end{aligned}$$

矩阵形式:
$$\begin{bmatrix} \dot{U}_1 \\ \dot{I}_1 \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} \dot{U}_2 \\ -\dot{I}_2 \end{bmatrix}$$

传输参数矩阵或A参数矩阵
$$\mathbf{A} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$$

互易性条件 $\Delta_A = A_{11}A_{22} - A_{12}A_{21} = 1$

对称条件 $\Delta_A = A_{11}A_{22} - A_{12}A_{21} = 1$ 及 $A_{11} = A_{22}$



§ 10.3 传输参数和混合参数

证明： 由：

$$\begin{aligned} \dot{I}_1 &= Y_{11}\dot{U}_1 + Y_{12}\dot{U}_2 \\ \dot{I}_2 &= Y_{21}\dot{U}_1 + Y_{22}\dot{U}_2 \end{aligned}$$

解得：

$$\begin{cases} \dot{U}_1 = -\frac{Y_{22}}{Y_{21}}\dot{U}_2 + \frac{1}{Y_{21}}\dot{I}_2 \\ \dot{I}_1 = (Y_{12} - \frac{Y_{11}Y_{22}}{Y_{21}})\dot{U}_2 + \frac{Y_{11}}{Y_{21}}\dot{I}_2 \end{cases}$$

$$\dot{U}_1 = A_{11}\dot{U}_2 + A_{12}(-\dot{I}_2)$$

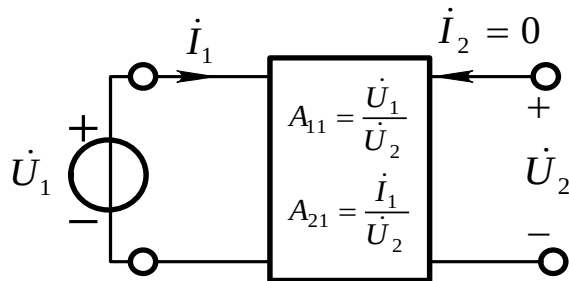
$$\dot{I}_1 = A_{21}\dot{U}_2 + A_{22}(-\dot{I}_2)$$

所以：

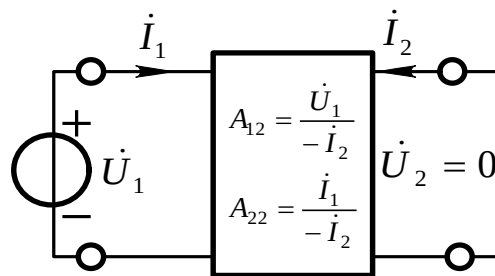
$$\Delta_A = A_{11}A_{22} - A_{12}A_{21} = \frac{Y_{22}Y_{11}}{Y_{21}^2} + \frac{Y_{12}Y_{21} - Y_{11}Y_{22}}{Y_{21}^2} = \frac{Y_{12}}{Y_{21}}$$

§ 10.3 传输参数和混合参数

传输参数的测定:



(a)



(b)

$$A_{11} = \left. \frac{\dot{U}_1}{\dot{U}_2} \right|_{\dot{I}_2 = 0} \quad A_{21} = \left. \frac{\dot{I}_1}{\dot{U}_2} \right|_{\dot{I}_2 = 0}$$

$$A_{12} = \left. \frac{\dot{U}_1}{-\dot{I}_2} \right|_{\dot{U}_2 = 0} \quad A_{22} = \left. \frac{\dot{I}_1}{-\dot{I}_2} \right|_{\dot{U}_2 = 0}$$

逆传输参数方程

$$\left. \begin{aligned} \dot{U}_2 &= B_{11}\dot{U}_1 - B_{12}\dot{I}_1 \\ \dot{I}_2 &= B_{21}\dot{U}_1 - B_{22}\dot{I}_1 \end{aligned} \right\}$$

矩阵形式:

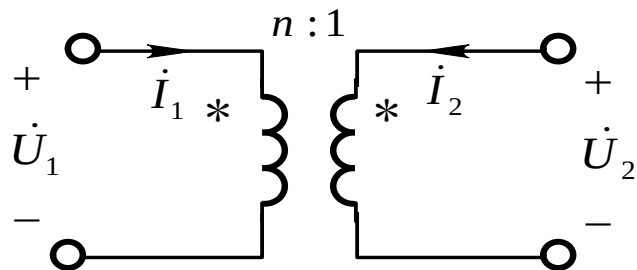
$$\begin{bmatrix} \dot{U}_2 \\ \dot{I}_2 \end{bmatrix} = \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix} \begin{bmatrix} \dot{U}_1 \\ -\dot{I}_1 \end{bmatrix}$$

逆传输参数矩阵

$$\mathbf{B} \neq \mathbf{A}^{-1}$$

$$\mathbf{B} = \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}$$

§ 10.3 传输参数和混合参数



理想变压器

左图表示二端口理想变压器。
根据它的元件方程写出传输参数方程：

$$\begin{bmatrix} \dot{U}_1 \\ \dot{I}_1 \end{bmatrix} = \begin{bmatrix} n & 0 \\ 0 & 1/n \end{bmatrix} \begin{bmatrix} \dot{U}_2 \\ -\dot{I}_2 \end{bmatrix}$$

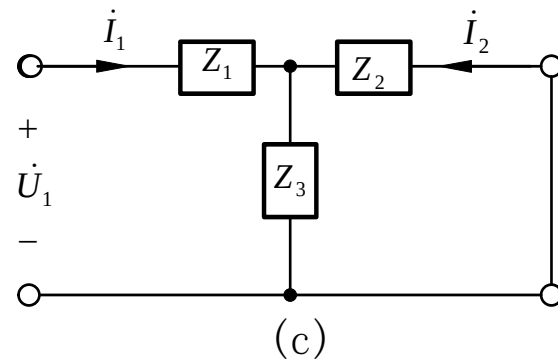
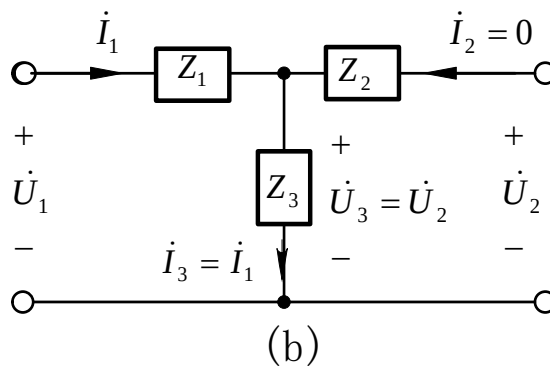
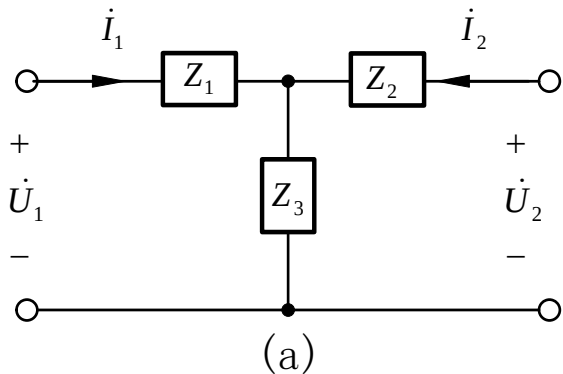
传输参数矩阵为：

$$\mathbf{A} = \begin{bmatrix} n & 0 \\ 0 & 1/n \end{bmatrix}$$

并且满足 $A_{11}A_{22} - A_{12}A_{21} = 1$ 的互易条件。由此方程可见，理想变压器不存在阻抗参数和导纳参数。

§ 10.3 传输参数和混合参数

例 求图(a)所示T形二端口网络的传输参数。



解

(1) 令 $i_2 = 0$ ，得图(b)由此求得 A_{11} 、 A_{21}

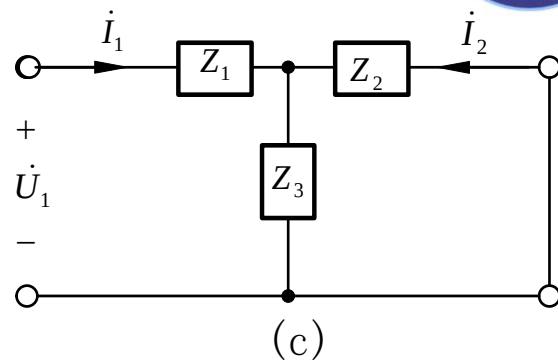
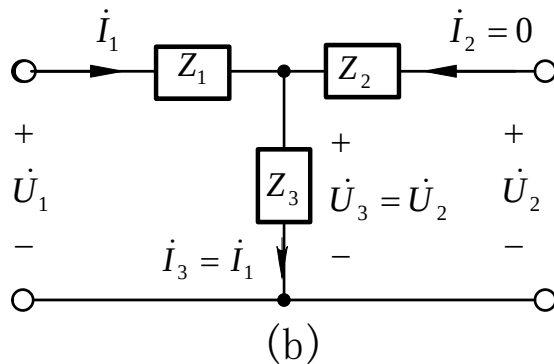
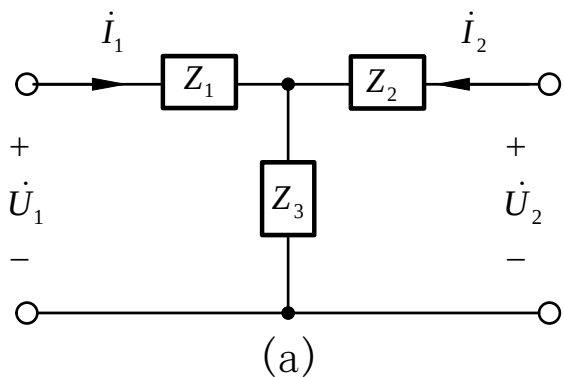
$$\dot{U}_2 = \dot{U}_3 = \frac{Z_3}{Z_1 + Z_3} \dot{U}_1 \quad A_{11} = \left. \frac{\dot{U}_1}{\dot{U}_2} \right|_{i_2=0} = 1 + \frac{Z_1}{Z_3}$$

$$\dot{I}_1 = \dot{I}_3 = \frac{\dot{U}_2}{Z_3} \quad A_{21} = \left. \frac{\dot{I}_1}{\dot{U}_2} \right|_{i_2=0} = \frac{1}{Z_3}$$

$$\begin{aligned} \dot{U}_1 &= A_{11}\dot{U}_2 + A_{12}(-\dot{I}_2) \\ \dot{I}_1 &= A_{21}\dot{U}_2 + A_{22}(-\dot{I}_2) \end{aligned}$$



§ 10.3 传输参数和混合参数



(2) 再令 $\dot{U}_2=0$ 得图 (c) 又求得 A_{12} 、 A_{22}

$$\dot{U}_1 = A_{11}\dot{U}_2 + A_{12}(-\dot{I}_2)$$

$$\dot{I}_1 = A_{21}\dot{U}_2 + A_{22}(-\dot{I}_2)$$

$$\dot{I}_2 = -\frac{Z_3}{Z_2 + Z_3}\dot{I}_1 \quad \dot{I}_1 = -(1 + \frac{Z_2}{Z_3})\dot{I}_2 \quad A_{22} = \left. \frac{\dot{I}_1}{-\dot{I}_2} \right|_{\dot{U}_2=0} = 1 + \frac{Z_2}{Z_3}$$

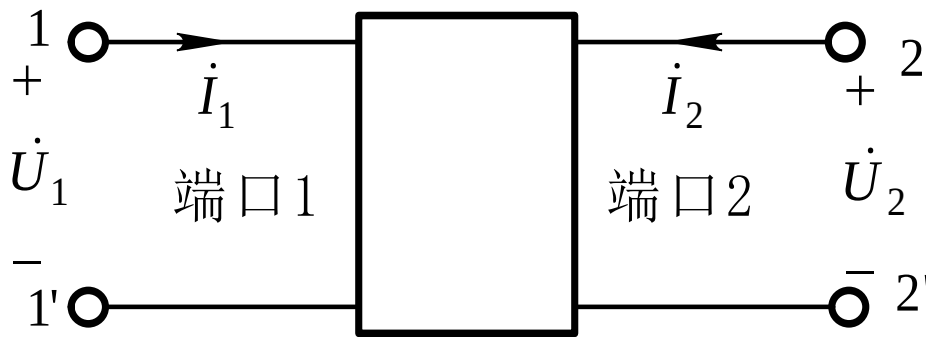
$$\dot{U}_1 = Z_1\dot{I}_1 - Z_2\dot{I}_2 = -Z_1(1 + \frac{Z_2}{Z_3})\dot{I}_2 - Z_2\dot{I}_2 = -(Z_1 + Z_2 + \frac{Z_1Z_2}{Z_3})\dot{I}_2$$

$$A_{12} = \left. \frac{\dot{U}_1}{-\dot{I}_2} \right|_{\dot{U}_2=0} = Z_1 + Z_2 + \frac{Z_1Z_2}{Z_3}$$



§ 10.3 传输参数和混合参数

2 混合参数方程



混合参数方程或 H 参数方程

$$\begin{aligned}\dot{U}_1 &= H_{11}\dot{I}_1 + H_{12}\dot{U}_2 \\ \dot{I}_2 &= H_{21}\dot{I}_1 + H_{22}\dot{U}_2\end{aligned}$$

矩阵形式:
$$\begin{bmatrix} \dot{U}_1 \\ \dot{I}_2 \end{bmatrix} = \begin{bmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{bmatrix} \begin{bmatrix} \dot{I}_1 \\ \dot{U}_2 \end{bmatrix}$$

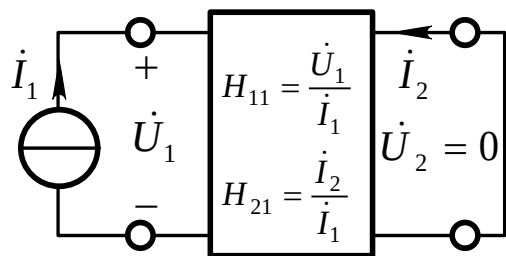
混合参数矩阵或 H 参数矩阵 $\mathbf{H} = \begin{bmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{bmatrix}$

互易性条件: $H_{12} = -H_{21}$

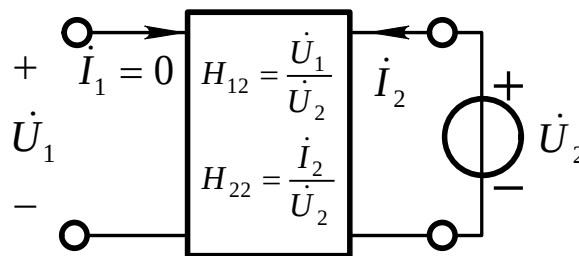
对称条件: $\Delta_H = H_{11}H_{22} - H_{12}H_{21} = H_{11}H_{22} + H_{12}^2 = 1$

§ 10.3 传输参数和混合参数

H 参数测定:



(a)



(b)

$$\begin{aligned} \dot{U}_1 &= H_{11}\dot{I}_1 + H_{12}\dot{U}_2 \\ \dot{I}_2 &= H_{21}\dot{I}_1 + H_{22}\dot{U}_2 \end{aligned}$$

$$H_{11} = \left. \frac{\dot{U}_1}{\dot{I}_1} \right|_{\dot{U}_2 = 0} \quad H_{21} = \left. \frac{\dot{I}_2}{\dot{I}_1} \right|_{\dot{U}_2 = 0} \quad H_{12} = \left. \frac{\dot{U}_1}{\dot{U}_2} \right|_{\dot{I}_1 = 0} \quad H_{22} = \left. \frac{\dot{I}_2}{\dot{U}_2} \right|_{\dot{I}_1 = 0}$$

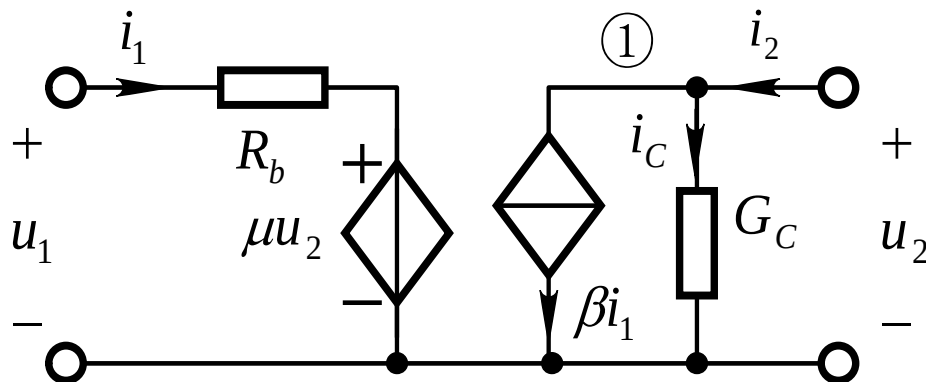
逆混合参数方程

$$\left. \begin{aligned} \dot{I}_1 &= G_{11}\dot{U}_1 + G_{12}\dot{I}_2 \\ \dot{U}_2 &= G_{21}\dot{U}_1 + G_{22}\dot{I}_2 \end{aligned} \right\} \quad \begin{bmatrix} \dot{I}_1 \\ \dot{U}_2 \end{bmatrix} = \begin{bmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{bmatrix} \begin{bmatrix} \dot{U}_1 \\ \dot{I}_2 \end{bmatrix} \quad \mathbf{G} = \begin{bmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{bmatrix}$$

G 和 H 互为逆矩阵 $\mathbf{G} = \mathbf{H}^{-1}$ 或 $\mathbf{H} = \mathbf{G}^{-1}$

§ 10.3 传输参数和混合参数

例 求图示半导体晶体管低频小信号等效电路的混合参数矩阵。



解 对输入端口所在回路列KVL方程 $u_1 = R_b i_1 + \mu u_2$

节点①列KCL方程: $i_2 = \beta i_1 + i_C = \beta i_1 + G_C u_2$

混合参数矩阵:
$$\mathbf{H} = \begin{bmatrix} R_b & \mu \\ \beta & G_C \end{bmatrix}$$

§ 10.4 二端口网络的等效电路

1. 互易二端口网络的等效电路

(1) 给定Z参数宜选用T形等效电路

对图(a)列回路电流方程

$$\left. \begin{aligned} \dot{U}_1 &= (Z_1 + Z_3)\dot{I}_1 + Z_3\dot{I}_2 \\ \dot{U}_2 &= Z_3\dot{I}_1 + (Z_2 + Z_3)\dot{I}_2 \end{aligned} \right\}$$

$$\mathbf{Z} = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix}$$

$$Z_{11} = Z_1 + Z_3 \quad Z_{12} = Z_{21} = Z_3 \quad Z_{22} = Z_2 + Z_3$$

$$\Rightarrow Z_1 = Z_{11} - Z_{12} \quad Z_2 = Z_{22} - Z_{12} \quad Z_3 = Z_{12}$$

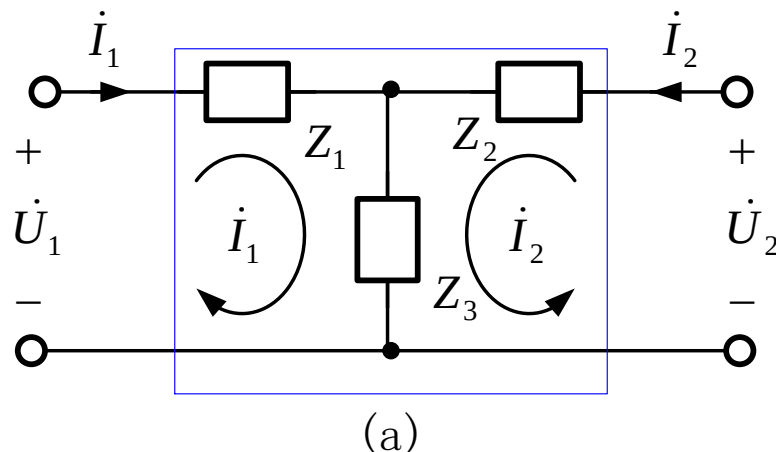


图14.20 二端口网络的T形等效电路

§ 10.4 二端口网络的等效电路

(2) 给定Y参数宜选Π形等效电路

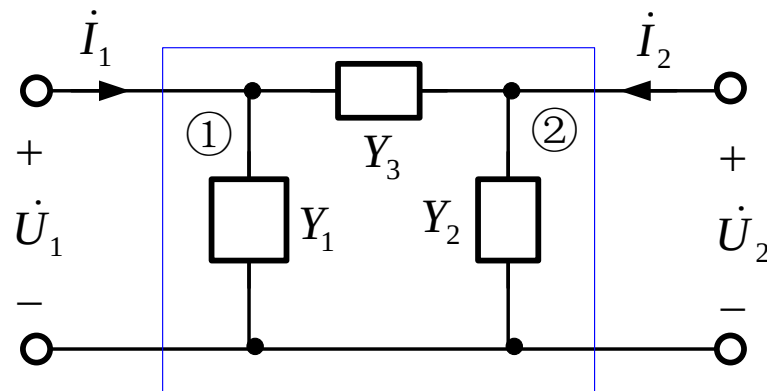
对图(b)列节点电压方程:

$$\left. \begin{aligned} \dot{I}_1 &= (Y_1 + Y_3)\dot{U}_1 - Y_3\dot{U}_2 \\ \dot{I}_2 &= -Y_3\dot{U}_1 + (Y_2 + Y_3)\dot{U}_2 \end{aligned} \right\}$$

$$\mathbf{Y} = \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix}$$

$$Y_{11} = Y_1 + Y_3 \quad Y_{12} = Y_{21} = -Y_3 \quad Y_{22} = Y_2 + Y_3$$

$$\Rightarrow \quad Y_1 = Y_{11} + Y_{12} \quad Y_2 = Y_{22} + Y_{21} \quad Y_3 = -Y_{12}$$



(b)

二端口网络的Π形等效电路

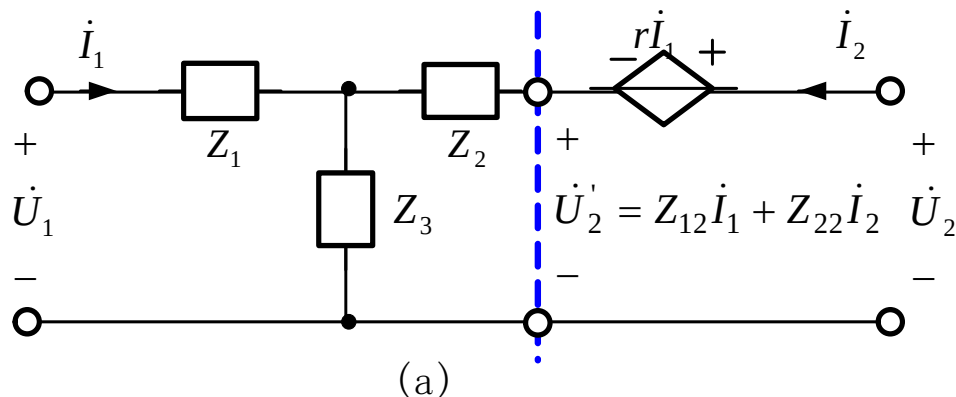
§ 10.4 二端口网络的等效电路

2. 非互易二端口网络的等效电路

(1) 给定Z参数

$$\left. \begin{aligned} \dot{U}_1 &= Z_{11}\dot{I}_1 + Z_{12}\dot{I}_2 \\ \dot{U}_2 &= Z_{21}\dot{I}_1 + Z_{22}\dot{I}_2 \end{aligned} \right\} \quad \begin{aligned} \dot{U}_1 &= Z_{11}\dot{I}_1 + Z_{12}\dot{I}_2 \\ \dot{U}_2 &= Z_{12}\dot{I}_1 + Z_{22}\dot{I}_2 + (Z_{21} - Z_{12})\dot{I}_1 \end{aligned}$$

虚线左侧仍是一个互易性二端口的表达式，可用T形电路来代替；而在虚线右侧部分，则是一个电流控制电压源。



非互易性二端口的 T 形等效电路

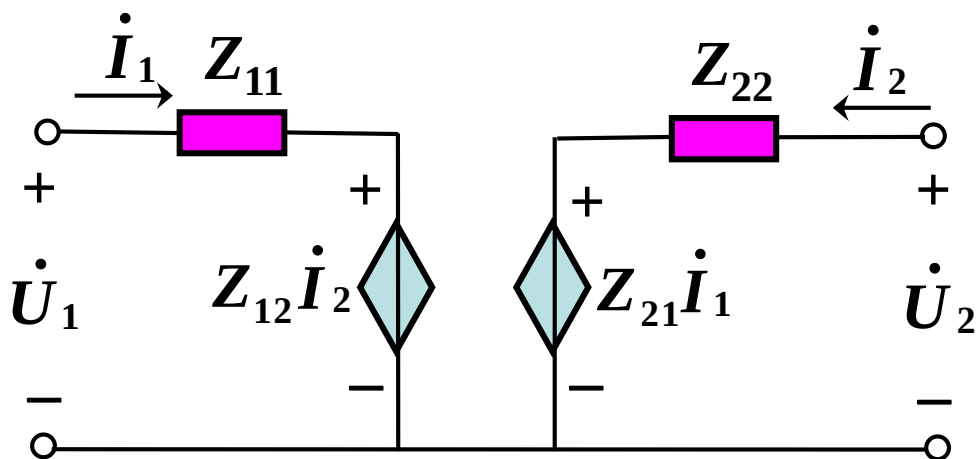
§ 10.4 二端口网络的等效电路

方法2: 直接由参数方程得到等效电路。

$$\dot{U}_1 = Z_{11}\dot{I}_1 + Z_{12}\dot{I}_2$$

$$\dot{U}_2 = Z_{21}\dot{I}_1 + Z_{22}\dot{I}_2$$

等效电路为:



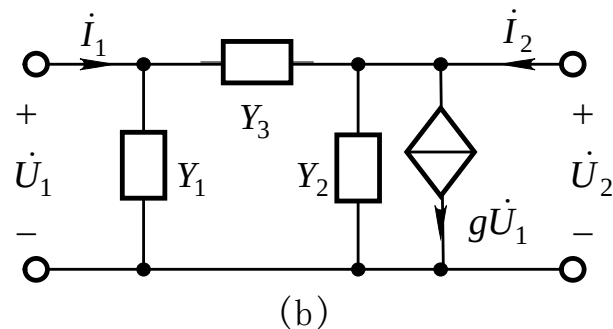
§ 10.4 二端口网络的等效电路

(2) 给定Y参数

$$\dot{I}_1 = Y_{11}\dot{U}_1 + Y_{12}\dot{U}_2$$

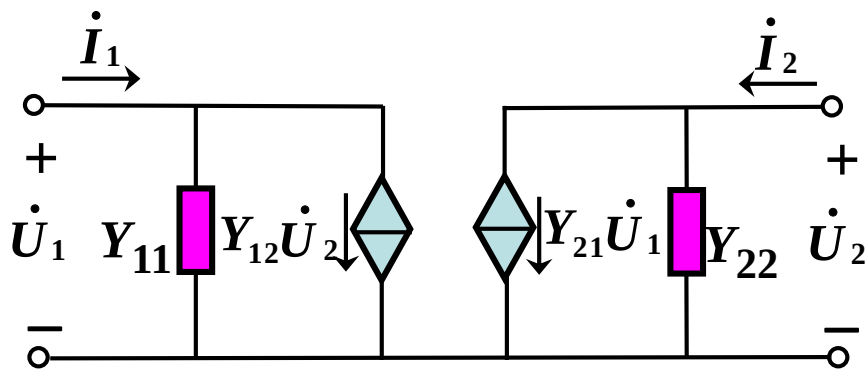
$$\dot{I}_2 = Y_{21}\dot{U}_1 + Y_{22}\dot{U}_2$$

$$\dot{I}_2 = Y_{12}\dot{U}_1 + Y_{22}\dot{U}_2 + (Y_{21} - Y_{12})\dot{U}_1$$



非互易性二端口的 Π 形等效电路

方法2: 直接由参数方程得到等效电路。



§ 10.4 二端口网络的等效电路

例 设二端口网络复频域导纳参数

矩阵为:

$$\mathbf{Y} = \begin{bmatrix} 2s+1 & -1 \\ -1.3 & 1+2/s \end{bmatrix} \text{ (单位 S)}$$

求它的 Π 形等效电路参数。

解 矩阵 $\mathbf{Y}$ 中 $Y_{12} \neq Y_{21}$ ，应该用含有受控源的电路来等效，如图(a)所示。求得

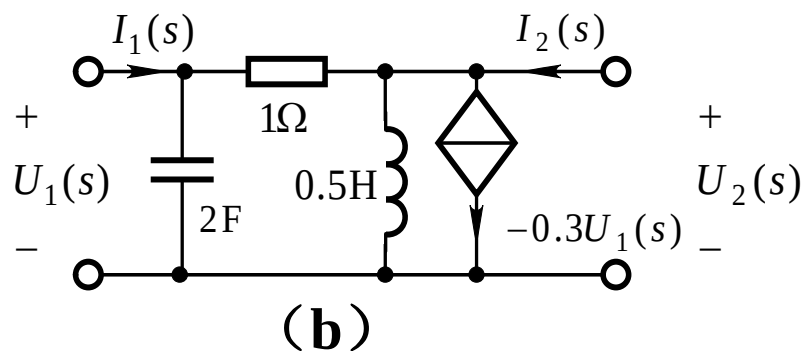
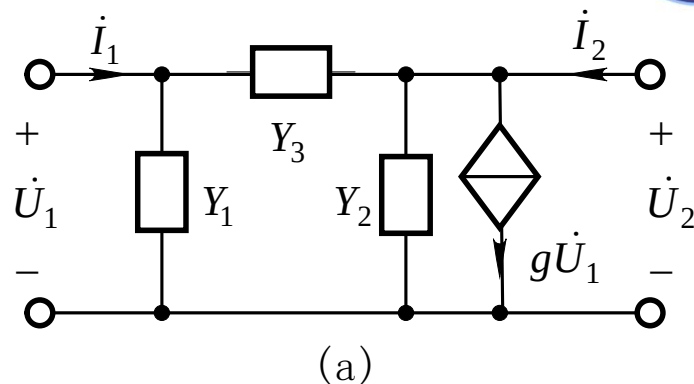
$$Y_1(s) = Y_{11}(s) + Y_{12}(s) = 2s$$

$$Y_2(s) = Y_{22}(s) + Y_{12}(s) = 2/s$$

$$Y_3(s) = -Y_{12}(s) = 1$$

$$g = Y_{21}(s) - Y_{12}(s) = -0.3$$

根据上述复频域中的导纳值可以得出对应的元件性质及参数值，如图(b)所示。

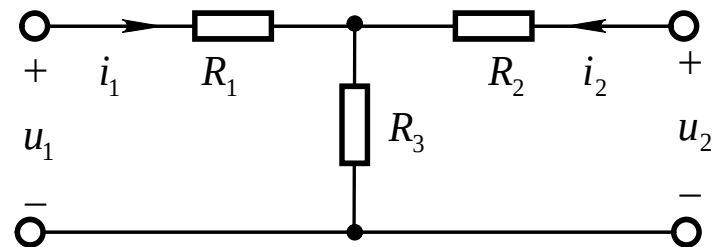


§ 10.4 二端口网络的等效电路

例 已知二端口网络的传输参数矩阵为

$$A = \begin{bmatrix} 1.3 & 13.4\Omega \\ 0.1S & 1.8 \end{bmatrix}$$

求它的T形等效电路。



解

$$\because A_{11}A_{22} - A_{12}A_{21} = 1$$

A满足互易条件，因此其T形等效电路不含受控源。

求T形电路的传输参数。

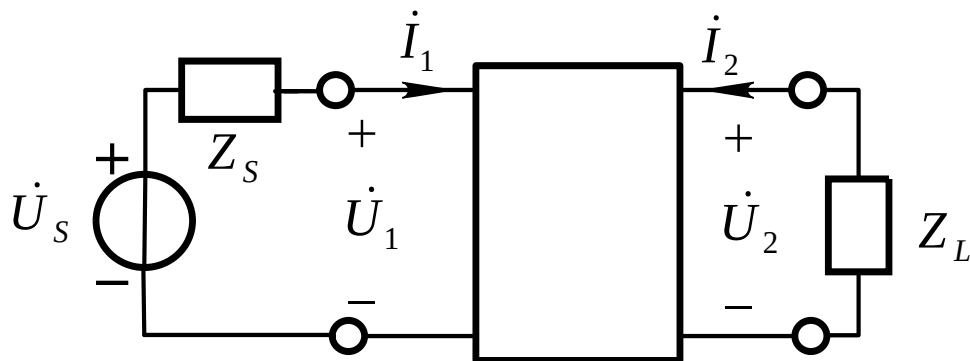
$$A_{11} = \left. \frac{\dot{U}_1}{\dot{U}_2} \right|_{i_2=0} = 1 + \frac{Z_1}{Z_3} \quad A_{21} = \left. \frac{\dot{I}_1}{\dot{U}_2} \right|_{i_2=0} = \frac{1}{Z_3} \quad A_{22} = \left. \frac{\dot{I}_1}{-\dot{I}_2} \right|_{\dot{U}_2=0} = 1 + \frac{Z_2}{Z_3}$$

$$\Rightarrow A_{11} = 1 + \frac{R_1}{R_3} = 1.3 \quad A_{21} = \frac{1}{R_3} = 0.1S \quad A_{22} = 1 + \frac{R_2}{R_3} = 1.8$$

由上述关系求得等效电路各电阻： $R_1 = 3\Omega$ $R_2 = 8\Omega$ $R_3 = 10\Omega$

§ 10.5 二端口网络与电源及负载的联接

1 二端口网络与电源和负载的联接



二端口网络与电源和负载的联接

约束端口的方程有：

(1) 二端口参数方程

$$\dot{U}_1 = A_{11}\dot{U}_2 + A_{12}(-\dot{I}_2)$$

(2) 电源支路方程

$$\dot{I}_1 = A_{21}\dot{U}_2 + A_{22}(-\dot{I}_2)$$

(3) 负载支路方程

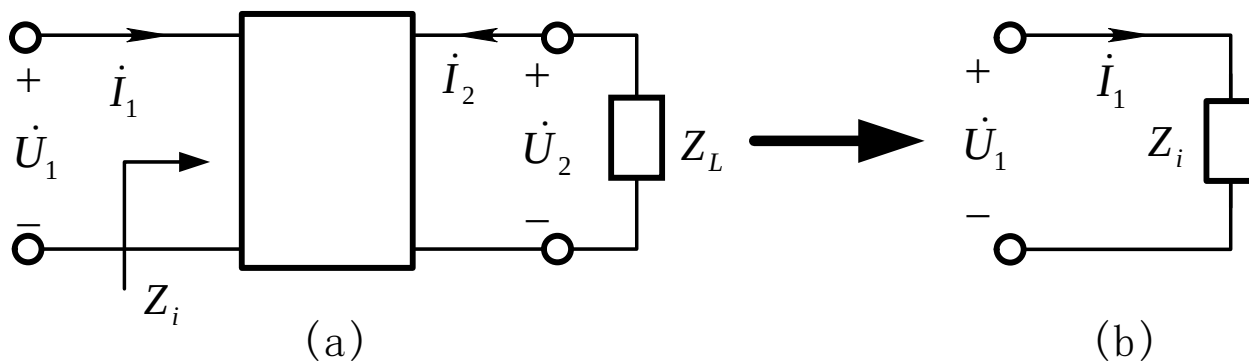
$$\dot{U}_1 = \dot{U}_s - Z_s \dot{I}_1$$

$$\dot{U}_2 = -Z_L \dot{I}_2$$

§ 10.5 二端口网络与电源及负载的联接

2 输入阻抗、输出阻抗

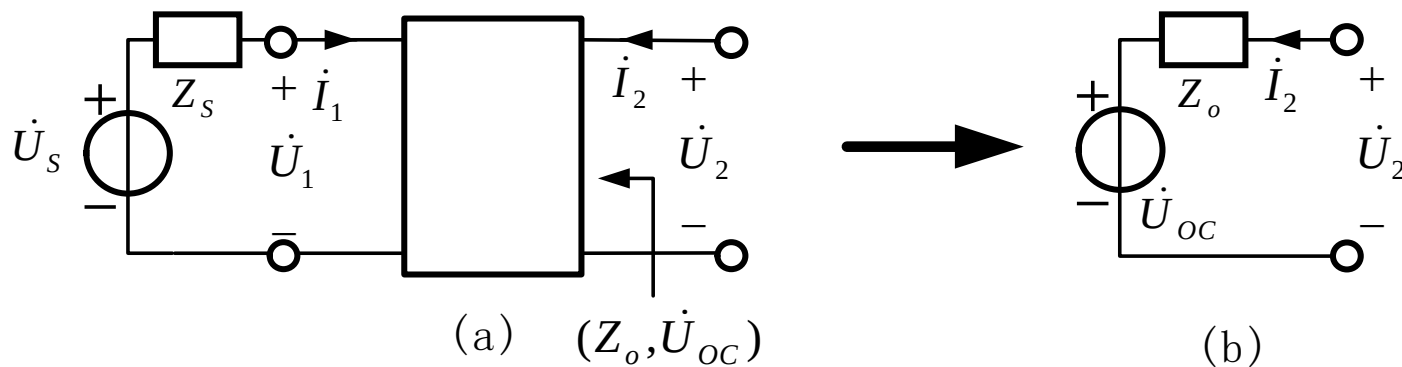
输入阻抗：从输入端口向二端口视入，是一个线性无独立源一端口网络，可用一个阻抗来等效代替，称为输入阻抗。



$$Z_i = \frac{\dot{U}_1}{\dot{I}_1} = \frac{A_{11}\dot{U}_2 - A_{12}\dot{I}_2}{A_{21}\dot{U}_2 - A_{22}\dot{I}_2} = \frac{A_{11}(-Z_L\dot{I}_2) - A_{12}\dot{I}_2}{A_{21}(-Z_L\dot{I}_2) - A_{22}\dot{I}_2} = \frac{A_{11}Z_L + A_{12}}{A_{21}Z_L + A_{22}}$$

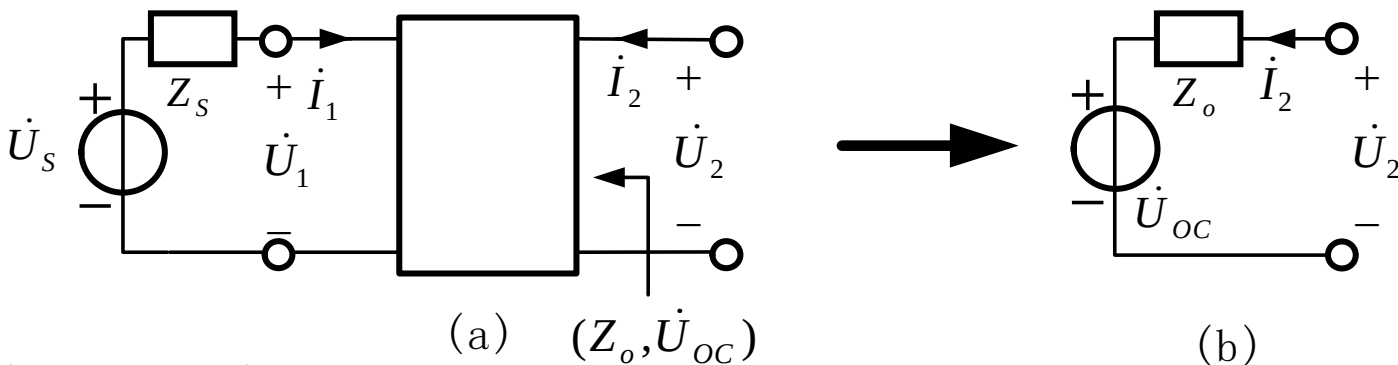
§ 10.5 二端口网络与电源及负载的联接

输出阻抗：从输出端口向二端口视入，是线性含独立源一端
口网络，可用戴维南电路等效代替。从输出端口视入的等效
阻抗，称为输出阻抗。



§ 10.5 二端口网络与电源及负载的联接

计算输出端口的戴维南等效电路：



$$\left. \begin{aligned} \dot{U}_1 &= A_{11}\dot{U}_2 + A_{12}(-\dot{I}_2) \\ \dot{I}_1 &= A_{21}\dot{U}_2 + A_{22}(-\dot{I}_2) \end{aligned} \right\} \Rightarrow A_{11}\dot{U}_2 - A_{12}\dot{I}_2 = \dot{U}_s - Z_s\dot{I}_1 = \dot{U}_s - Z_s(A_{21}\dot{U}_2 - A_{22}\dot{I}_2)$$

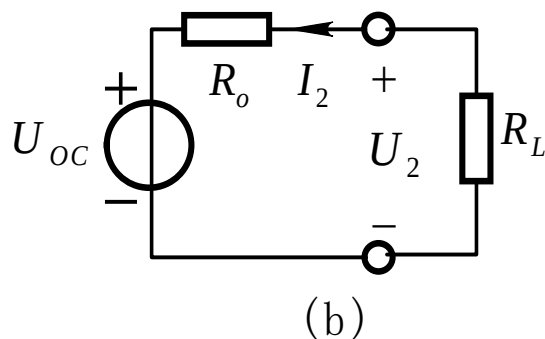
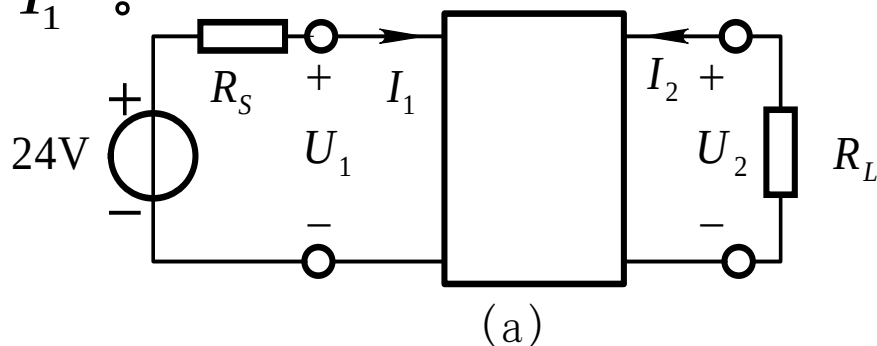
$$\dot{U}_1 = \dot{U}_s - Z_s\dot{I}_1 \Rightarrow \dot{U}_2 = \frac{\dot{U}_s}{A_{21}Z_s + A_{11}} + \frac{A_{22}Z_s + A_{12}}{A_{21}Z_s + A_{11}}\dot{I}_2 = \dot{U}_{OC} + Z_o\dot{I}_2$$

开路电压与等效输出阻抗

$$\dot{U}_{OC} = \frac{\dot{U}_s}{A_{21}Z_s + A_{11}} \quad Z_o = \left. \frac{\dot{U}_2}{\dot{I}_2} \right|_{\dot{U}_s=0} = \frac{A_{22}Z_s + A_{12}}{A_{21}Z_s + A_{11}}$$

§ 10.5 二端口网络与电源及负载的联接

例 下图所示电路的传输参数矩阵为 $\mathbf{A} = \begin{bmatrix} 1.5 & 5\Omega \\ 0.25\text{S} & 1.5 \end{bmatrix}$ $R_S = 2\Omega$
求负载电阻 R_L 消耗的最大功率以及此时输入端口的电压 U_1 和电流 I_1 。



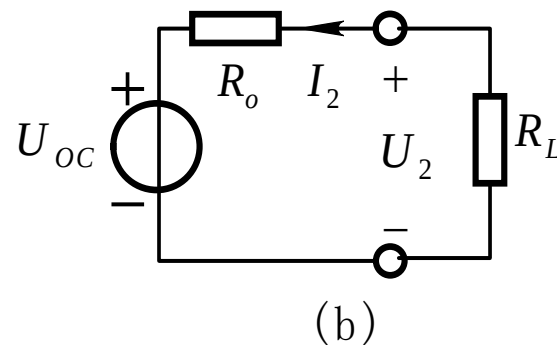
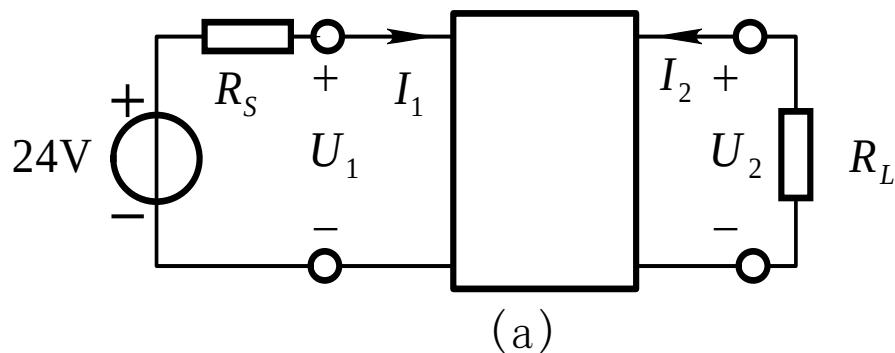
解

1) 求输出端口的戴维南等效电路

$$U_{OC} = \frac{U_S}{A_{21}R_S + A_{11}} = \frac{24\text{V}}{0.25 \times 2 + 1.5} = 12\text{V}$$

$$R_0 = \frac{U_2}{I_2} = \frac{A_{22}R_S + A_{12}}{A_{21}R_S + A_{11}} = \frac{(1.5 \times 2 + 5)\Omega}{0.25 \times 2 + 1.5} = 4\Omega$$

§ 10.5 二端口网络与电源及负载的联接



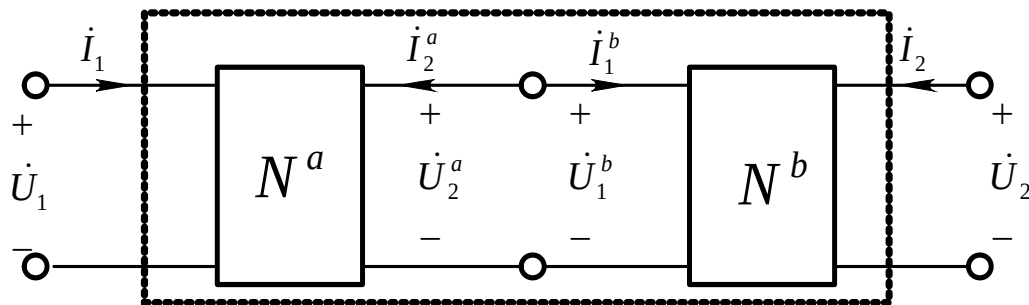
2) 根据最大功率传输定理: $R_L = R_o = 4\Omega$ 时, 吸收功率最大

$$P_{\max} = \frac{U_{OC}^2}{4R_o} = \frac{(12V)^2}{4 \times 4\Omega} = 9W$$

$$U_2 = \frac{1}{2}U_{OC} = 6V \quad I_2 = -\frac{U_{OC}}{2R_o} = -1.5A$$

$$\begin{bmatrix} U_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} U_2 \\ -I_2 \end{bmatrix} = \begin{bmatrix} 1.5 & 5\Omega \\ 0.25S & 1.5 \end{bmatrix} \begin{bmatrix} 6V \\ 1.5A \end{bmatrix} = \begin{bmatrix} 16.5V \\ 3.75A \end{bmatrix}$$

§ 10.6 二端口网络的级联



二端口网络的级联

已知:
$$\begin{bmatrix} \dot{U}_1 \\ \dot{I}_1 \end{bmatrix} = \begin{bmatrix} A_{11}^a & A_{12}^a \\ A_{21}^a & A_{22}^a \end{bmatrix} \begin{bmatrix} \dot{U}_2^a \\ -\dot{I}_2^a \end{bmatrix}$$

$$\begin{bmatrix} \dot{U}_1^b \\ \dot{I}_1^b \end{bmatrix} = \begin{bmatrix} A_{11}^b & A_{12}^b \\ A_{21}^b & A_{22}^b \end{bmatrix} \begin{bmatrix} \dot{U}_2 \\ -\dot{I}_2 \end{bmatrix}$$

N^a 输出端口与 N^b 输入端口存在
$$\begin{bmatrix} \dot{U}_2^a \\ -\dot{I}_2^a \end{bmatrix} = \begin{bmatrix} \dot{U}_1^b \\ \dot{I}_1^b \end{bmatrix}$$

可得
$$\begin{bmatrix} \dot{U}_1 \\ \dot{I}_1 \end{bmatrix} = \begin{bmatrix} A_{11}^a & A_{12}^a \\ A_{21}^a & A_{22}^a \end{bmatrix} \begin{bmatrix} A_{11}^b & A_{12}^b \\ A_{21}^b & A_{22}^b \end{bmatrix} \begin{bmatrix} \dot{U}_2 \\ -\dot{I}_2 \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} \dot{U}_2 \\ -\dot{I}_2 \end{bmatrix}$$

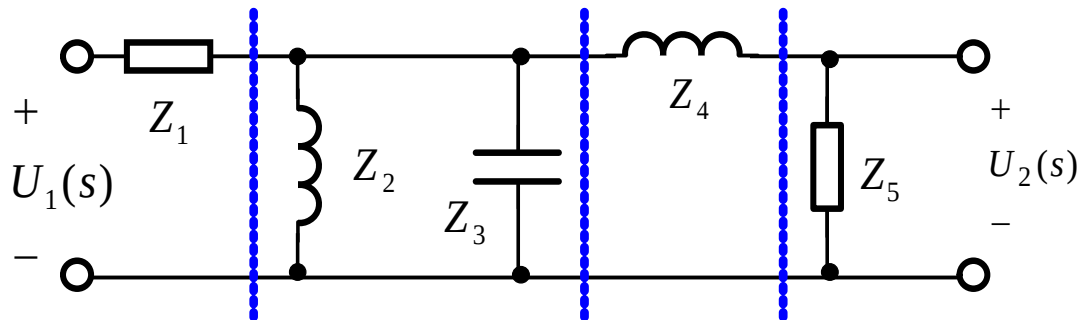
$$\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} = \begin{bmatrix} A_{11}^a & A_{12}^a \\ A_{21}^a & A_{22}^a \end{bmatrix} \begin{bmatrix} A_{11}^b & A_{12}^b \\ A_{21}^b & A_{22}^b \end{bmatrix}$$

可推广到多个二端口级联的情况。注意，矩阵相乘的顺序应与级联顺序一致

§ 10.6 二端口网络的级联

例 求下图所示二端口的传输参数, 并求输出端口开路时转移函数

$U_2(s)/U_1(s)$. $Z_1(s) = 1\Omega$, $Z_2(s) = 2s$, $Z_3(s) = 1/(3s)$, $Z_4(s) = 4s$, $Z_5(s) = 1$



解

将电路看成由四个二端口级联组成, 各级的传输参数矩阵为

$$\begin{bmatrix} A_{11}^{(1)} & A_{12}^{(1)} \\ A_{21}^{(1)} & A_{22}^{(1)} \end{bmatrix} = \begin{bmatrix} 1 & Z_1(s) \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1\Omega \\ 0 & 1 \end{bmatrix} \begin{bmatrix} A_{11}^{(2)} & A_{12}^{(2)} \\ A_{21}^{(2)} & A_{22}^{(2)} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ \frac{Z_2(s) + Z_3(s)}{Z_2(s)Z_3(s)} & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ \frac{6s^2 + 1}{2s} & 1 \end{bmatrix}$$

$$\begin{bmatrix} A_{11}^{(3)} & A_{12}^{(3)} \\ A_{21}^{(3)} & A_{22}^{(3)} \end{bmatrix} = \begin{bmatrix} 1 & 4s \\ 0 & 1 \end{bmatrix} \quad \begin{bmatrix} A_{11}^{(4)} & A_{12}^{(4)} \\ A_{21}^{(4)} & A_{22}^{(4)} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ \frac{1}{Z_5(s)} & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1S & 1 \end{bmatrix}$$



§ 10.6 二端口网络的级联

整个二端口网络传输参数矩阵为

$$\begin{aligned} \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} &= \begin{bmatrix} 1 & 1\Omega \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ \frac{6s^2+1}{2s} & 1 \end{bmatrix} \begin{bmatrix} 1 & 4s \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1S & 1 \end{bmatrix} \\ &= \begin{bmatrix} \frac{24s^3+14s^2+8s+1}{2s} & (12s^2+4s+3) \\ \frac{24s^3+6s^2+6s+1}{2s} & 12s^2+3 \end{bmatrix} \end{aligned}$$

没接负载时由二端口网络传输参数方程 $\dot{U}_1 = A_{11}\dot{U}_2 + A_{12}(-\dot{I}_2)$

$$\dot{I}_1 = A_{21}\dot{U}_2 + A_{22}(-\dot{I}_2)$$

可得: $U_1(s) = A_{11}U_2(s) - A_{12}I_2(s) = A_{11}U_2(s)$

$$\text{求得转移电压比为 } K_U(s) = \left. \frac{U_2(s)}{U_1(s)} \right|_{I_2=0} = \frac{1}{A_{11}} = \frac{2s}{24s^3+14s^2+8s+1}$$



小结

- ❖ 本章首先介绍二端口网络的各种参数方程及两种重要等效电路：T形和 Π 形等效电路；
- ❖ 其次介绍二端口网络的输入、输出阻抗等概念；
- ❖ 最后介绍二端口网络的级联规律。